

PROJECT & INFRASTRUCTURE REPORT

Bullion Infrastructure

A real, CI/CD-deployed multi-cloud Kubernetes platform running AWS EKS and GCP GKE as **one active-active cluster** — joined by an Istio service mesh, with live traffic load-balanced across both clouds.

EKS + GKE

Istio multi-primary

Active-active LB

GitHub Actions + OIDC

Terraform IaC

✓ Active-active cross-cloud mesh LIVE & verified — traffic balances EKS ⇄ GKE (12 AWS / 8 GCP)

Repo: Olisaemeka111/bullion-infrastructure · AWS acct 730767193290 · GCP project-f38bfcd6 · Report 2026-06-23

AWS / EKS

GCP / GKE

Service mesh

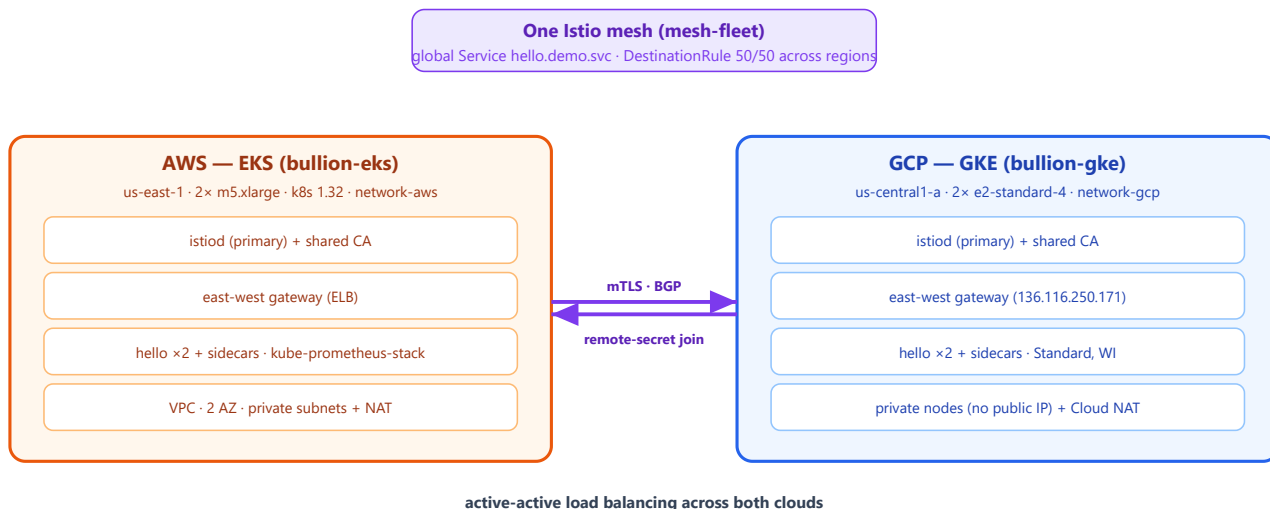
CI/CD

Observability

Pending

1 Goal & architecture

Run clusters in more than one cloud and have them behave as **one active-active platform** — not primary + cold backup. A workload deployed to both clouds is one global service; requests load-balance across both at once; if a cloud fails, traffic auto-redistributes to the survivor. All infrastructure-as-code + CI/CD, keyless auth, gated approvals.



2 Cloud inventory (final state)

AWS · US-EAST-1 EKS — bullion-eks

Cluster	EKS, Kubernetes 1.32
Nodes	2 × m5.xlarge (4 vCPU/16GB)
Network	VPC, 2 AZ, private subnets + 1 NAT, public subnets
Access	EKS access entries (CI role + Olisa = admin); node SG open for istiod 15017/15012
State	S3 <code>fleet-mini-tfstate-59998ac4</code> , native locking

GCP · US-CENTRAL1-A GKE — bullion-gke

Cluster	GKE Standard (zonal), Workload Identity
Nodes	2 × e2-standard-4 (4 vCPU/16GB)
Network	VPC-native; private nodes + Cloud NAT egress
Why private	free-trial IP quota (4) — keeps node IPs off the count for the mesh LBs
Project	<code>project-f38bfcd6-70ff-4a8d-9cd</code>

Azure — disabled. The account has no subscription (`NoAzurePlanFound`); the azurerm provider + AKS module are commented out until one exists. Re-enabling is a documented toggle, no redesign.

3 CI/CD & security

All deploys run through **GitHub Actions** with **keyless OIDC** auth and a gated `production` environment (required reviewer). No static cloud keys.

WORKFLOW	PURPOSE	TRIGGER
<code>ci.yml</code>	80 stdlib tests + <code>terraform fmt/validate</code> (no creds)	every push / PR
<code>deploy.yml</code>	Terraform: EKS, GKE, cross-cloud VPN — plan → gated apply → destroy	PR / push / dispatch
<code>platform.yml</code>	Istio multi-primary join + global app + observability	dispatch / mesh changes

Keyless cloud auth

AWS: GitHub OIDC → IAM role `fleet-mini-gha`. **GCP:** OIDC → Workload Identity Federation → SA `fleet-mini-gha`. No long-lived keys.

State & secrets

S3 **native state locking** (no DynamoDB). Cloud auth is keyless OIDC; only the non-sensitive OIDC identifiers are stored, as **GitHub Actions secrets/variables** (set via `iac/generate-credentials.sh` using the `gh` CLI).

4 Service mesh — final state

The two clusters form **one mesh** (`meshID = mesh-fleet`) via Istio **multi-primary, multi-network**.

How they're joined

- **Shared root CA** → workloads in EKS & GKE trust each other's mTLS.
- **A primary istiod per cluster** — no control-plane SPOF.
- **East-west gateways** (LB) carry cross-cluster mTLS traffic.
- **Remote secrets** exchanged → each istiod discovers the other's endpoints.
- **Network labels** `topology.istio.io/network` (`network-aws` / `network-gcp`) — the key to cross-network discovery.

Global service & LB

`hello.demo.svc.cluster.local` runs in both clusters. A `DestinationRule` sets an explicit **50/50 cross-region split** with `outlierDetection` so a failed cloud is ejected and traffic redistributes (active-active failover).

East-west gateways: EKS `...elb.amazonaws.com` · GKE `136.116.250.171`.

✓ Verified active-active behaviour

From a client pod in **EKS**, 20 calls to the global service returned **aws: 12 / gcp: 8** — traffic from one cloud is served by **both** clouds through the joined mesh.

```
# from an EKS pod
for i in $(seq 20); do curl -s hello.demo.svc.cluster.local; done
hello from aws    x12
hello from gcp    x8
```

5 Tasks completed

#	TASK	STATUS
1	Tooling (gh, terraform, aws, gcloud, az, kubectl, istioctl)	✓
2	Cloud identities via OIDC (AWS role, GCP WIF)	✓
3	Remote state (S3 native locking)	✓
4	CI/CD pipelines + gated production env	✓
5	Provision EKS (VPC, node group)	✓
6	Provision GKE (Standard, private nodes)	✓
7	Local kubectl access to both clusters	✓
8	Istio multi-primary install on both	✓
9	Join clusters (CA + east-west + remote secrets + network labels)	✓
10	Global app + active-active 50/50 load balancing	✓ verified
11	Observability (Prometheus/Grafana on EKS)	✓
12	Cross-cloud VPN (AWS↔GCP HA VPN + BGP)	✓ live (4 tunnels)
13	CockroachDB central DB (multi-region over VPN, via CI/CD)	✓ live — 4 nodes
14	North-south ingress gateways (public LB :80)	✓ live
15	Board game — global active-active app (GHCR → both clouds)	✓ live
16	Control-plane container (agentic reconciler + dashboard)	pending
17	Global DNS (single-endpoint north-south LB)	module built (optional)

Notable blockers resolved

SYMPTOM	FIX
EKS rejected K8s 1.29	bump to 1.32
GKE Autopilot can't host self-managed Istio / DaemonSets	switch to GKE Standard
Node pool rejected <code>GKE_METADATA</code>	enable Workload Identity on cluster
istiod won't schedule on medium nodes	4 vCPU/16GB nodes (m5.xlarge / e2-standard-4)
GCP free-trial IP quota (4) exceeded	private nodes + Cloud NAT
Terraform state drift (partial applies)	fresh plan reconciled
EKS sidecar-injector webhook timeout	open node SG :15017 / :15012
DestinationRule distribute+failover rejected	distribute + outlierDetection
Cross-cluster traffic stayed local	add <code>topology.istio.io/network</code> namespace label

6 Deployed vs pending · operate · costs

✓ Live now

EKS + GKE; Istio multi-primary mesh (joined); global app with active-active cross-cloud LB; **cross-cloud VPN** (4 tunnels); central **CockroachDB** (4 nodes, via CI/CD); north-south ingress gateways + the **board game** served from both clouds; observability on EKS; full CI/CD with OIDC + gated approvals.

⌚ Pending (optional)

In-cluster control-plane container; single global DNS endpoint (north-south); production hardening (CockroachDB TLS, least-privilege IAM, private GKE endpoint).

Operate

```
# access
aws eks update-kubeconfig --name bullion-eks --region us-east-1 --alias eks
gcloud container clusters get-credentials bullion-gke --zone us-central1-a --project project-f38bfcd6-70ff-4a8d-9cd
# deploy: PR → plan, merge → gated apply ; teardown: Actions → deploy → action=destroy
```

Cost & security: both clusters + LoadBalancers bill while running — run the `destroy` workflow when idle. Rotate the bootstrap AWS key (shared in plaintext during setup). The `Olisa` user holds AdministratorAccess (scope down beyond testing); GKE control-plane endpoint is public for CI/local access.

Bullion Infrastructure · active-active multi-cloud Kubernetes · repo Olisaemeka111/bullion-infrastructure · report generated 2026-06-23. — End of report.